

Graphing Radical Functions

A **radical function** of the form $f(x) = a\sqrt[n]{x-h} + k$, where n is an integer greater than 1, is in a family of functions whose parent function is $f(x) = \sqrt[n]{x}$.

Graphing a Square Root Function

If $n = 2$, the family contains square root functions.

EXAMPLE 1

Graph $f(x) = \sqrt{x}$.

SOLUTION

Use a table to find several ordered pairs of the function. Include only nonnegative values for x because $\sqrt{x}$ is undefined if x is negative. For convenience, choose perfect squares for x .

► **Think About It** Every positive real number has two square roots.

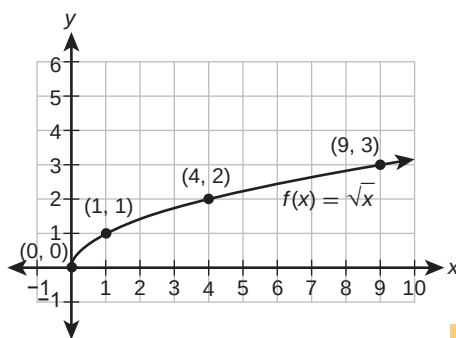
For example, the two square roots of 9 are 3 and -3 . But the $\sqrt{}$ symbol indicates only the principal (nonnegative) square root. Therefore, $\sqrt{9} = 3$.

x	0	1	4	9	16
$f(x)$	0	1	2	3	4

domain: $\{x \mid x \geq 0\}$

range: $\{y \mid y \geq 0\}$

The graph looks like the top half of a sideways parabola.



Graphing a Cube Root Function

If $n = 3$, the family contains cube root functions.

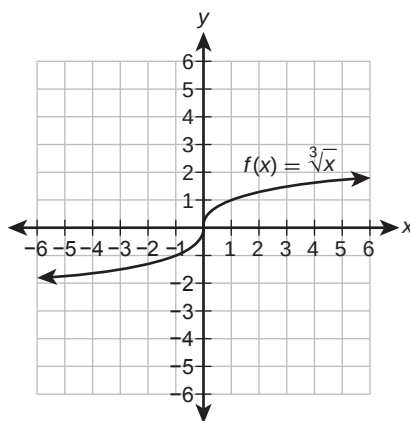
EXAMPLE 2

Graph $f(x) = \sqrt[3]{x}$.

SOLUTION

Use a table to find several ordered pairs of the function. Negative numbers have cube roots, so include negative values for x . For convenience, choose integer values for x that have integer cube roots.

x	-27	-8	-1	0	1	8	27
$f(x)$	-3	-2	-1	0	1	2	3



domain: $\{x \mid x \in \mathbb{R}\}$

range: $\{y \mid y \in \mathbb{R}\}$ ■

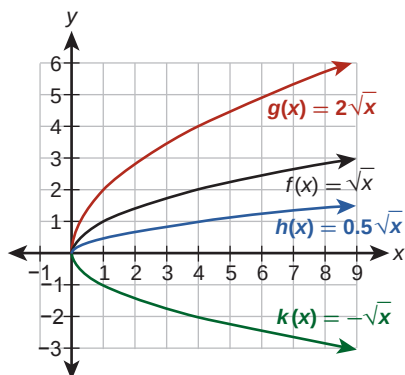
Graphing Families of n th Root Functions

Every radical function of the form $f(x) = a\sqrt[n]{x-h} + k$ is a transformation of a parent function $f(x) = \sqrt[n]{x}$.

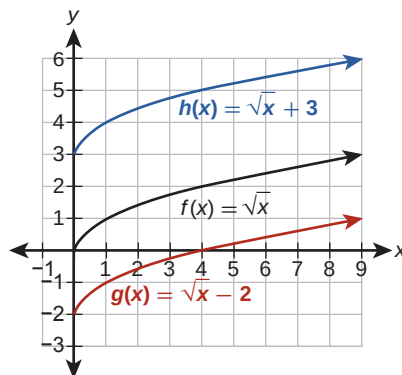
Square Root Functions

Square Root Function Graph Family: $f(x) = a\sqrt{x-h} + k$

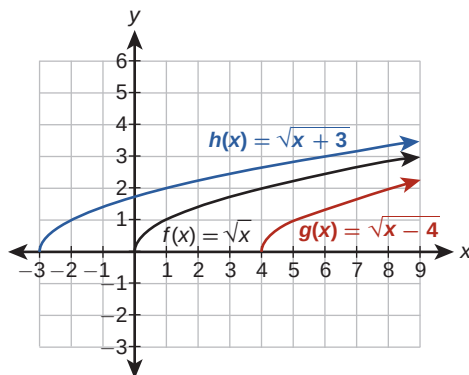
Each change in the value of a , h , or k results in a different transformation of the parent graph, $f(x) = \sqrt{x}$.



Changing only the parameter a causes the parent function to reflect, stretch, or compress.



Changing only the parameter k causes the parent function to shift up or down on the y-axis.

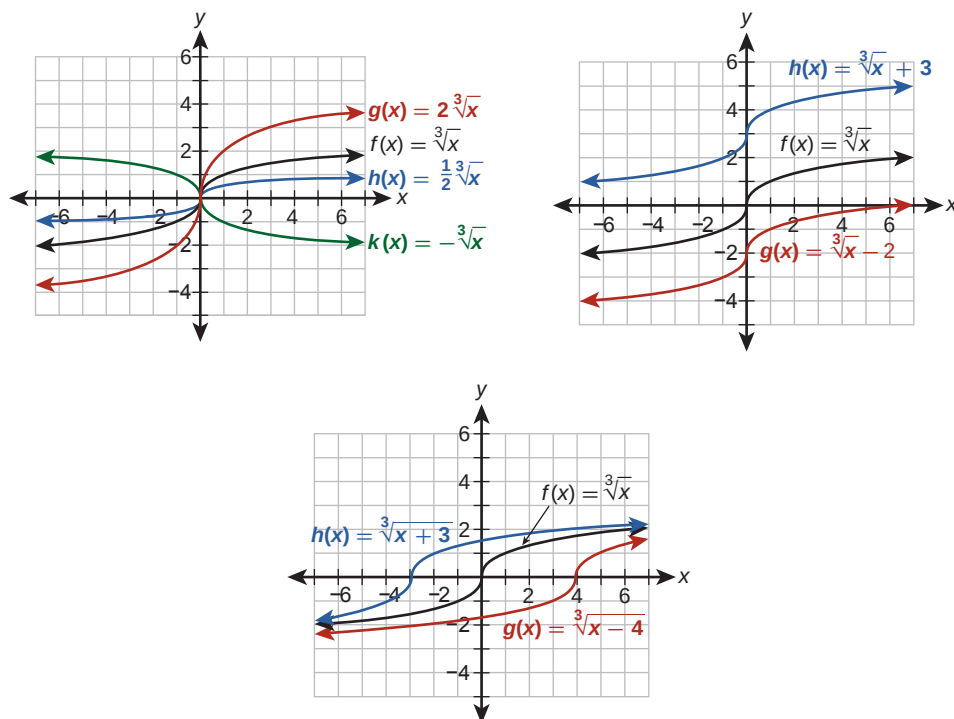


Changing only the parameter h causes the parent function to shift left or right on the x-axis.

Cube Root Functions

Cube Root Function Graph Family: $f(x) = a\sqrt[3]{x-h} + k$

Each change in the value of a , h , or k results in a different transformation of the parent graph, $f(x) = \sqrt[3]{x}$.



All graphs of functions with even roots look similar, and all graphs of functions with odd roots look similar. The only way to tell the graphs apart is to look at specific ordered pairs.

To transform the graph of $f(x) = \sqrt[n]{x}$ by changing more than one parameter, first decide whether or not to reflect it across the x -axis. Then apply each transformation to the graph obtained from the previous transformation.

Graphing a Square Root Function and Identifying the Domain and Range

EXAMPLE 3

For the function $g(x) = -3\sqrt{x+1} - 2$:

A Graph the function.

SOLUTION

Step 1 Describe how to obtain the graph by transforming the graph of $f(x) = \sqrt{x}$.

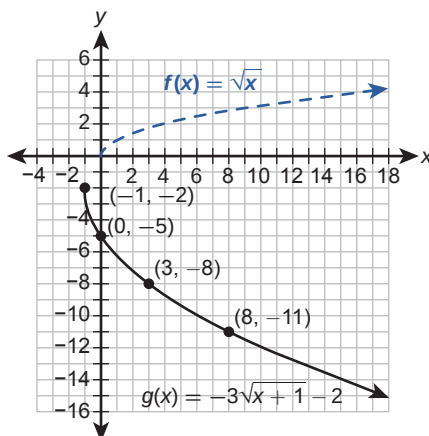
$g(x) = -3\sqrt{x+1} - 2$ is in the form $g(x) = a\sqrt{x-h} + k$, with $a = -3$, $h = -1$, and $k = -2$.

Parameter	Transformation
$a < 0$	Reflect the graph of $f(x) = \sqrt{x}$ across the x -axis.
$a = -3$	Stretch vertically by a factor of $ -3 = 3$.
$h = -1$	Translate 1 unit to the left.
$k = -2$	Translate 2 units down.

Step 2 Make a table of values. Then plot points and sketch the graph. If you choose x -values so that $x+1$ is a perfect square, then calculating corresponding y -values becomes easy. For example,

$$g(8) = -3\sqrt{8+1} - 2 = -3\sqrt{9} - 2 = -3 \cdot 3 - 2 = -11.$$

x	$g(x)$
-1	-2
0	-5
3	-8
8	-11



► **Think About It** You might prefer to plot points and sketch the graph first, and then use the transformation description to check that your graph is reasonable.

B Identify the function's domain and range.

SOLUTION

domain: The radicand is $x + 1$. The radicand in a square root expression must be nonnegative, so solve the inequality $x + 1 \geq 0$. The solution set of $x + 1 \geq 0$ is given by $x \geq -1$. The domain, then, is $\{x \mid x \geq -1\}$.

range: Since $\sqrt{x + 1} \geq 0$, $-3\sqrt{x + 1} \leq 0$ and $-3\sqrt{x + 1} - 2 \leq -2$.

The range is $\{y \mid y \leq -2\}$. ■

Verify that the domain and range are reasonable, based on the graph.

► **Remember** The radicand is the expression under the radical symbol.

Graphing a Cube Root Function and Identifying the Domain and Range

EXAMPLE 4

For the function $g(x) = \frac{1}{2}\sqrt[3]{x - 1} + 3$:

A Graph the function.

SOLUTION

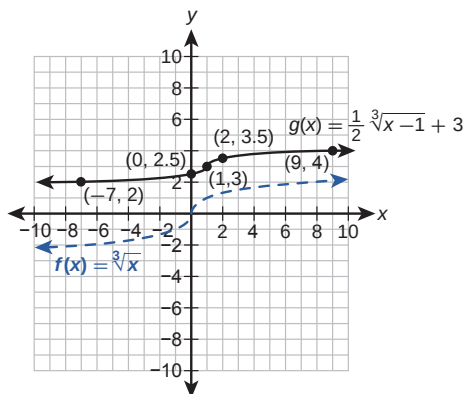
Step 1 Describe how to obtain the graph by transforming the graph of $f(x) = \sqrt[3]{x}$.

$g(x) = \frac{1}{2}\sqrt[3]{x - 1} + 3$ is in the form $g(x) = a\sqrt[3]{x - h} + k$, with $a = \frac{1}{2}$, $h = 1$, and $k = 3$.

Parameter	Transformation
$a = \frac{1}{2}$	Compress the graph of $f(x) = \sqrt[3]{x}$ vertically by a factor of $\frac{1}{2}$.
$h = 1$	Translate 1 unit to the right.
$k = 3$	Translate 3 units up.

Step 2 Make a table of values. Then plot points and sketch the graph.

x	$g(x)$
-7	2
0	2.5
1	3
2	3.5
9	4



► **Think About It** If you choose x -values for your table that make $x - 1$ an integer whose cube root is an integer, then corresponding function values are easy to calculate. For example,

$$\begin{aligned}
 g(-7) &= \frac{1}{2} \sqrt[3]{-7-1} + 3 \\
 &= \frac{1}{2} \sqrt[3]{-8} + 3 \\
 &= \frac{1}{2} \cdot (-2) + 3 = 2
 \end{aligned}$$

B Identify the function's domain and range.

SOLUTION

domain: The index of the radical is 3, an odd integer. So the radicand $x - 1$ can be positive, negative, or zero. $\sqrt[3]{x-1}$ is defined for all real values of x . The domain is $\{x \mid x \in \mathbb{R}\}$.

range: Since x can be any real number, all of the following expressions can be any real number: $x - 1$, $\sqrt[3]{x-1}$, $\frac{1}{2} \sqrt[3]{x-1}$, and $\frac{1}{2} \sqrt[3]{x-1} + 3$. The range is $\{y \mid y \in \mathbb{R}\}$. ■